

■ Cases

`Cases[{ $e_1$ ,  $e_2$ , ...}, pattern]` gives a list of the e_i which match the pattern.

`Cases[{ $e_i$ , ...}, rhs, pattern]` gives a list of the values of *rhs* corresponding to the e_i which match the pattern.

Examples: `Cases[{5, k[3], 2, k[1], k[2]}, k[_]]` $\longrightarrow$ {k[3], k[1], k[2]}.

■ `Cases[{5, k[3], k[5]}, k[x_], 4 x + 1]` $\longrightarrow$ {13, 21}. ■ The first argument to `Cases` need not have head `List`. ■ See page 235. ■ See also: `Select`, `Position`.